

WEEK 40 POPULATION SUBDIVISION, DEMOGRAPHY INFERENCE AND LD

BOOK EXERCISES



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EVOLUTIONARY THINKING
29. AUGUST 2025

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PRIMATE OF THE WEEK

Philippine Tarsier

Carlito syrichta

Intermediate between lemurs and monkeys

One of the smallest primates on earth

Native to southeastern Philippines

Because of their spines, they can turn their head nearly 180 degrees

→ because their eyes are so big that they cannot move their eyes in the sockets

If human eyes were as proportionally large as tarsiers', they would be as big as grapefruits

Nocturnal and arboreal

Mainly use olfactory communication instead of vocal

Babies are born with open eyes



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PLAN FOR TODAY

We will focus on **chapter 4 and 5** → Population structure

You will focus on chapter 6 Friday → Linkage disequilibrium

1. Little recap of the main concepts

2. In your own group

Make the book exercises of chapter 4, 5



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CHAPTER 4: POPULATION SUBDIVISION

Population structure = when the population is not randomly mating because of geographic structure

Chapter 4: effects of population structure **of genetic differentiation** and **methods for quantifying** such differentiation



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CHAPTER 4: POPULATION SUBDIVISION

- Wahlund effect: H_S vs H_T
- H_S : Heterozygosity within subpopulations
Compute $2pq$ in each population and then average = the heterozygosity you actually observe
- H_T : Heterozygosity in pooled population
Average allele frequencies first → then compute $2pq$, as if everyone mated randomly

$$H_t = H_s + \delta^2/2, \text{ where } \delta = |f_1 - f_2|$$

Different allele frequencies → $H_t > H_s$ → fewer heterozygotes than HW predicts. That deficit is the Wahlund effect.



IF 2 SUBPOPULATION HAVE THE SAME ALLELE FREQUENCIES. IS THERE A WAHLUND EFFECT?

No

$$\delta = |f_1 - f_2| = 0$$

→ so $H_t = H_s$.



F_{ST} QUANTIFIES THE SUBDIVISION

$$F_{ST} = (H_T - H_S) / H_T$$

The proportion of heterozygosity lost because the population is subdivided.

0 = identical frequencies

1 = different alleles fixed.

0 – 0.05

0.05 – 0.15

0.15 – 0.25

> 0.25

Little

Moderate

Great

Very great



MIGRATION FORWARDS AND BACKWARDS IN TIME

Forward: Wright-Fisher + migration

$$f_1(t+1) = (1 - m) f_1(t) + m f_2(t)$$

Migration pulls allele frequencies together. Drift, mutation and selection push them apart.

Island model: the island ends up with the mainland's frequency.

Backwards: coalescent + migration

$M = 2Nm$ migrations per generation

- each lineage migrates at rate M
- Each pair in the same population coalesces at rate 1
- lineages in different populations cannot coalesce

Total rate: $(i+j)*M + i(i-1)/2 + j(j-1)/2$



COALESCENCE TIMES GIVE F_{ST} UNDER MIGRATION MODEL

$$E[s] = 2$$

→ two lineages in the same populations. This is independent of M

$$E[d] = 1(2M) + 2$$

= the waiting time for migration + $E[s]$

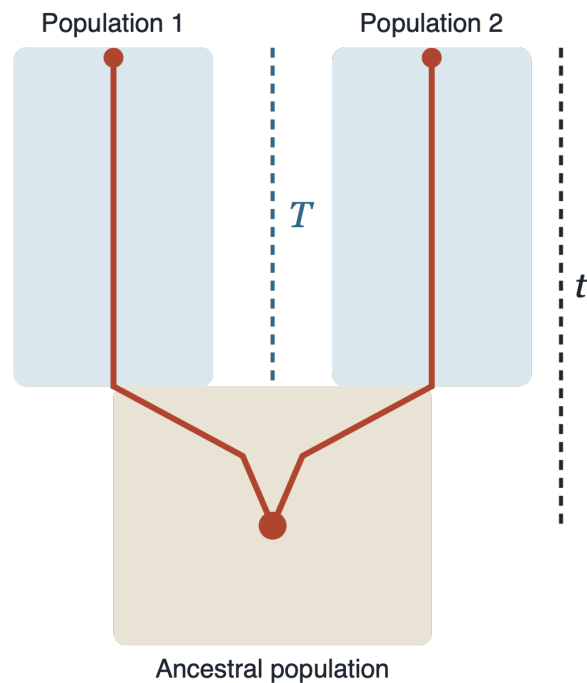
→ different populations. Small M means a long wait before they can meet

$F_{ST} = 1/(1+8M)$ → two populations and infinite sites.

→ from the observed F_{ST} you can calculate Migration rate



DIVERGENCE MODELS: SPLIT, THEN NO GENE FLOW



Quantity	Value
Same population	$E[s] = 1$
Different populations	$E[d] = T + 1$
Differentiation	$F_{st} = T / (T + 2)$

$t \geq T$: dating a split from the MRCA of two sequences overestimates it.

Same F_{st} , two stories: any F_{st} value fits both a migration and a divergence model.



CHAPTER 5: INFERRING POPULATION HISTORY

Statistic: any thing you can calculate from the data (# of segregating sites, or F_{st})
→ you then plug the statistic into a model → solve for the parameter you want

→ You then loose a lot of information
→ that's why can you also estimate a tree, to retain more information
(all kinds of different trees, as you know : parsimony, distance, maximum likelihood)

A gene tree $\neq$ species tree

→ $t > T$ → so gene trees overestimate split times

→ **incomplete lineage sorting** → several lineages survive back into the ancestral population and may coalesce in the "wrong" order, making the gene tree disagree with the species tree.



GENOME WIDE DATA

- **Recombination**

→ every region has its own tree

- **Site frequency spectrum**

→ population growth gives long external branches and extra singletons → fitting the SFS can give information about splits, migration rates and population sizes

- **Admixture**

→ an individual can have ancestry from several populations and the analysis estimates what fraction comes from each



EXERCISES

Chapter 4: 4.1-4.7

Exercise 4.8-4.9 are proofs -> you DO NOT have to do this

Chapter 5: 5.1-5.5

You will work on [chapter 6 Linkage equilibrium on Friday!](#)



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